APM5557: Partial Differential Equations
Homework 2 Problems

Frank Palazzolo

Problem 1. From problem 4.1-3 in the book

Prove that the solution of the Robin or third boundary value problem for the Laplace equation:
$\Delta u = 0$ in Ω, with $\frac{\partial u}{\partial n} + \alpha u = \beta$ on $\partial\Omega$
is unique when $\alpha > 0$ is a constant.

Solution. Suppose u_1 and u_2 are two solutions to the Robin boundary value problem. Then their difference $v = u_1 - u_2$ satisfies the Laplace equation and the homogeneous Robin boundary condition:

$\Delta v = 0$ in Ω and $\frac{\partial v}{\partial n} + \alpha v = 0$ on $\partial\Omega$.

Now we use Green's first identity for v and v, which states that for a sufficiently smooth function v on a domain Ω with boundary $\partial\Omega$:

$$\int_\Omega \left(v\Delta v + |\nabla v|^2 \right) dx = \int_{\partial\Omega} v \, \frac{\partial v}{\partial n} \, dS. \tag{1}$$

Now, since $\Delta v = 0$, the left-hand side simplifies to:

$$\int_\Omega |\nabla v|^2 \, dx = \int_{\partial\Omega} v \, \frac{\partial v}{\partial n} \, dS. \tag{2}$$

From the Robin boundary condition, we have $\frac{\partial v}{\partial n} = -\alpha v$ on $\partial\Omega$. Substituting this into the boundary integral gives:

$$\int_\Omega |\nabla v|^2 \, dx = \int_{\partial\Omega} v(-\alpha v) \, dS = -\alpha \int_{\partial\Omega} v^2 \, dS. \tag{3}$$

Since $\alpha > 0$, the right-hand side is non-positive, while the left-hand side is non-negative.

This means that $\nabla v = 0$ in Ω and $\alpha v = 0$ and $v = 0$ on $\partial\Omega$

Since $\nabla v = 0$ in Ω, v is constant in Ω, and since $v = 0$ on the boundary, it must be that $v = 0$ everywhere in Ω. □

Problem 2. From problem 4.1-7a in the book

If Ω is a bounded domain and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfies $\Delta u = 0$ in Ω, then $\max_{\overline{\Omega}} |u| = \max_{\partial\Omega} |u|$.

Solution. Solution goes here. □

Problem 3. From problem 4.1-7b in the book

If $\Omega = \{x \in \mathbb{R}^n : |x| > 1\}$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfies $\Delta u = 0$ in Ω and $\lim_{|x|\to\infty} u(x) = 0$, then $\max_{\overline{\Omega}} |u| = \max_{\partial\Omega} |u|$.

Solution. Solution goes here. □

Problem 4. From problem 4.1-9a in the book

Suppose $u \in C(\Omega)$ satisfies the mean value property in Ω: for $\xi \in \Omega$ and $r > 0$ with $\overline{B_r(\xi)} = \{x : |x - \xi| \leq r\} \subset \Omega$,

$$u(\xi) = \frac{1}{\omega_n} \int_{|x|=1} u(\xi + rx)\, dS_x, \tag{4}$$

where ω_n is the measure of the $(n-1)$-dimensional sphere in $\mathbb{R}^n$. Show that u satisfies the maximum principle: either u is a constant, or

$$u(\xi) < \sup_{x \in \Omega} u(x) \quad \text{for all } \xi \in \Omega. \tag{5}$$

Solution. Solution goes here. □

Problem 5. From problem 4.2-4 in the book

(a) Use the weak maximum principle

$$\max_{x \in \overline{\Omega}} u(x) = \max_{x \in \partial\Omega} u(x) \tag{6}$$

to prove that the Green's function satisfies $G(x, \xi) \leq 0$ for $x, \xi \in \Omega$ with $x \neq \xi$.

(b) Use the strong maximum principle (5) to prove that $G(x, \xi) < 0$ for $x, \xi \in \Omega$ with $x \neq \xi$.

Solution. Solution goes here. □

Problem 6. From problem 4.2-7 in the book

If $u \in C(\Omega)$ satisfies the mean value property (4), then u is harmonic in Ω.

Solution. Solution goes here. □